

second project which is calculation for slope stability

1-Determine the factor of safety against sliding of the embankment with the slices method "simplified Bishop". you should consider 3 different slip circles

- base failure,
- slope failure with positive tangent
- slope failure with negative tangent at the slope toe

- Determine the factor of safety against sliding for the cut with the block method.

for the drawing part:

you should draw a horizontal surface then measure your height then make the road inclination

.draw the slops then find the ground surface intersections after that draw the trenches

2-The sliding soil mass is divided vertically in cells (slices)

: 8-10 slices depending on geometry, maximum 1,5 m wide slices

mine was less than 1.5 m width and there was 10 slices for more precision

3-Curved sliding surface can be considered a straight line at the bottom of the slice

♦ The inclination α, n is measured

4-For the analysis of the overall stability of a slope and any other geotechnical structure, the design approach DA-3 shall be used, along with the sets A2 „+“ M2 „+“ R1 of the partial factors.

Soil parameters	Symbol	Set	
		M1	M2 – Slopes ^b
Angle of shearing resistance ^a	$\gamma_{\phi'}$	1,0	1,35
Effective cohesion	$\gamma_{c'}$	1,0	1,35
Undrained shear strength	γ_{cu}	1,0	1,5
Unconfined strength	γ_{qu}	1,0	1,5
Unit weight	γ_{γ}	1,0	1,0

5-Tangent at the intersection with the crest should not curved backwards (maximum vertical)!

✓ The intersection at the toe (case a) also your decision!